Grade 7 - Module 4 Comprehensive Test - TEACHER ANSWER KEY

Scheduled: 2027-02-19 | Covers all four module weeks and all five subjects

Math: 7.G.1-5 | ELA: W.7.3; RL.7.5-7; L.7.1-6 | Science: P.7.5C-E | Social Studies: 7.5 Ancient Rome | Spanish: MS World Languages Framework 2016 - Novice communication

Award points by item. Accept equivalent correct mathematical forms and supported written interpretations. For open responses, use the stated rubric; do not require identical wording. Score each subject separately, then apply your grading policy.

Math - 28 points

Score: _____ / 28

1. At scale 1 cm : 6 m, what actual length is 5 cm? (1 point)

Answer / scoring: 30 m

2. At scale 1 cm : 6 m, what drawing length represents 42 m? (1 point)

Answer / scoring: 7 cm

3. A 4 cm side becomes 12 cm. Scale factor? (1 point)

Answer / scoring: 3

4. A 2 by 3 rectangle is enlarged by factor 4. New dimensions? (1 point)

Answer / scoring: 8 by 12

5. How does its area change under scale factor 4? (1 point)

Answer / scoring: Multiplied by 16

6. A map uses 1 cm : 2 km. A route measures 6.5 cm. Find the distance and show a proportion or equation. (2 points)

Answer / scoring: 6.5 x 2 = 13 km; equivalent proportion 1/2 = 6.5/d.

7. Can side lengths 2, 3, and 6 form a triangle? (1 point)

Answer / scoring: No: 2 + 3 < 6.

8. Can side lengths 4, 5, and 6 form a triangle? (1 point)

Answer / scoring: Yes: each pair sums to more than the third side.

9. Draw a triangle with sides 3 cm, 4 cm, and 5 cm using ruler and compass. (1 point)

Answer / scoring: Lengths match (allow +/-2 mm); sides close into a triangle.

10. What is the angle sum of a triangle? (1 point)

Answer / scoring: 180

11. A triangle has angles 40 and 65 degrees. Find the third. (1 point)

Answer / scoring: 75

12. Explain why 3, 3, and 7 cannot form a triangle. (2 points)

Answer / scoring: The two shorter sides sum to 6, less than 7; they cannot meet to form a closed triangle.

13. An angle is 35 degrees. Find its complement. (1 point)

Answer / scoring: 55

14. An angle is 120 degrees. Find its supplement. (1 point)

Answer / scoring: 60

15. One of two vertical angles is 68 degrees. Find the other. (1 point)

Answer / scoring: 68

16. A straight angle is split into 2x degrees and 50 degrees. Find x. (1 point)

Answer / scoring: 65

17. A triangle has angles 45 and 75 degrees. Find the third. (1 point)

Answer / scoring: 60

18. Adjacent angles on a straight line measure (3x + 10) and (2x + 20) degrees. Write and solve an equation. (2 points)

Answer / scoring: 5x + 30 = 180; x = 30. Angles 100 and 80 degrees.

19. A circle has radius 4 cm. Give its diameter. (1 point)

Answer / scoring: 8 cm

20. A circle has radius 4 cm. Give circumference in terms of pi. (1 point)

Answer / scoring: 8pi cm

21. A circle has radius 4 cm. Give area in terms of pi. (1 point)

Answer / scoring: 16pi square cm

22. A circle has diameter 10 cm. Using pi = 3.14, find circumference. (1 point)

Answer / scoring: 31.4 cm

23. A circle has radius 3 cm. Using pi = 3.14, find area. (1 point)

Answer / scoring: 28.26 square cm

24. Explain why the circumference formula uses diameter while area uses the square of radius. State both formulas and find each for radius 4. (2 points)

Answer / scoring: C = pi d = 2pi r = 8pi; A = pi r^2 = 16pi. Circumference is length; area measures a two-dimensional region.

ELA - 16 points

Score: _____ / 16

The Bridge Apprentice

In this original historical fiction, an apprentice watched an experienced builder inspect a Roman-style stone bridge. The apprentice wanted to remove the wooden supports as soon as the final stone was placed. "A finished appearance is not enough," the builder warned. He checked the arch and explained how its stones pressed against one another and transferred forces toward the supports. The apprentice grew impatient, but a cracked stone changed her mind. Together they replaced it and repeated the inspection. Later, the apprentice wrote that the delay had been the most important part of the work. The scene does not establish facts about a particular ancient builder; it uses an imagined conversation to explore patience and responsibility. Readers should distinguish the fictional characters from historical claims, such as the Romans' use of arches in engineering.
Background for comparison: Hermes in Greek tradition and Mercury in Roman tradition are messengers associated with travel.

1. Identify a character or participant and a challenge in the passage. (1 point)

Answer / scoring: Accurate participant plus relevant problem; in Grade 7 Module 4, apprentice and unsafe/premature bridge work.

2. What does pacing mean in a narrative? (1 point)

Answer / scoring: The speed at which events are presented; detail/dialogue can slow important moments.

3. Punctuate: Maya said I found the book (1 point)

Answer / scoring: Maya said, "I found the book."

4. Punctuate: Where is it asked Leo (1 point)

Answer / scoring: "Where is it?" asked Leo.

5. What is one role shared by Hermes and Mercury? (1 point)

Answer / scoring: Messenger/travel; this comparison is supplied in Grade 4 Module 8. For other modules, use the background statement: Hermes in Greek tradition and Mercury in Roman tradition are messengers associated with travel.

6. Why can cultures have similar stories with different names? (1 point)

Answer / scoring: Stories and traditions can be adapted, shared, or interpreted differently; similarity alone does not prove identical origins.

7. Identify a character or participant and a challenge in the passage. (1 point)

Answer / scoring: Accurate participant plus relevant problem; in Grade 7 Module 4, apprentice and unsafe/premature bridge work.

8. What does pacing mean in a narrative? (1 point)

Answer / scoring: The speed at which events are presented; detail/dialogue can slow important moments.

9. Which event changes the apprentice's attitude? (1 point)

Answer / scoring: Finding a cracked stone.

10. Name one fictional element and one historical idea in the passage. (1 point)

Answer / scoring: Imagined apprentice/conversation; Romans used arches in engineering.

11. Correct this misspelling: diffrent. (1 point)

Answer / scoring: different

12. Write an original short narrative of two organized paragraphs (at least 8 sentences total) about solving a problem. Include a clear sequence, a setting, one descriptive detail, and an action or line of dialogue. Edit conventions. Include one reflection about how your interpretation might differ from another reader's. (5 points)

Answer / scoring: Rubric (5): clear sequence (1); character/setting (1); specific descriptive detail (1); meaningful action/dialogue (1); readable complete sentences and conventions (1). Accept varied original stories, not only the supplied passage.

Science - 28 points

Score: _____ / 28

W1 data: A fictional comparison records temperature before and after a controlled warming step. Trial A: 21 temperature in degrees C; Trial B: 29 temperature in degrees C. These are invented assessment data, not a claim about a real investigation.

W2 data: A fictional comparison records temperature before and after a controlled warming step. Trial A: 22 temperature in degrees C; Trial B: 30 temperature in degrees C. These are invented assessment data, not a claim about a real investigation.

W3 data: A fictional comparison records temperature before and after a controlled warming step. Trial A: 23 temperature in degrees C; Trial B: 31 temperature in degrees C. These are invented assessment data, not a claim about a real investigation.

W4 data: A fictional comparison records temperature before and after a controlled warming step. Trial A: 24 temperature in degrees C; Trial B: 32 temperature in degrees C. These are invented assessment data, not a claim about a real investigation.

1. Does a visible change always mean a chemical reaction? (1 point)

Answer / scoring: No; physical changes can also be visible.

2. Use W1 data. Which trial has the higher recorded value, and by how much? (2 points)

Answer / scoring: Trial B; 8 temperature in degrees C.

3. Explain how to investigate a possible chemical reaction. (3 points)

Answer / scoring: Rubric (3): scientifically accurate parts/ideas (1); correct relationships/reasoning (1); clear labels/evidence or method (1). Observe identity-related evidence, compare alternatives, and use multiple observations.

4. What does an indicator help reveal? (1 point)

Answer / scoring: A property such as pH through a color response; it does not identify every substance uniquely.

5. Use W2 data. Which trial has the higher recorded value, and by how much? (2 points)

Answer / scoring: Trial B; 8 temperature in degrees C.

6. Interpret a hypothetical pH chart: A=3, B=7, C=10. (3 points)

Answer / scoring: Rubric (3): scientifically accurate parts/ideas (1); correct relationships/reasoning (1); clear labels/evidence or method (1). A acidic, B neutral, C basic; classify without tasting or handling unknowns.

7. Why can a reaction's surroundings change temperature? (1 point)

Answer / scoring: Energy is transferred between the reaction system and surroundings.

8. Use W3 data. Which trial has the higher recorded value, and by how much? (2 points)

Answer / scoring: Trial B; 8 temperature in degrees C.

9. Draw an energy-transfer model for each reaction type. (3 points)

Answer / scoring: Rubric (3): scientifically accurate parts/ideas (1); correct relationships/reasoning (1); clear labels/evidence or method (1). Correct direction of energy arrows and the appropriate temperature tendency.

10. Why might measured mass fall in an open container? (1 point)

Answer / scoring: A gas or other material may escape beyond the measured boundary.

11. Use W4 data. Which trial has the higher recorded value, and by how much? (2 points)

Answer / scoring: Trial B; 8 temperature in degrees C.

12. Model a closed reaction with equal atom counts before and after. (3 points)

Answer / scoring: Rubric (3): scientifically accurate parts/ideas (1); correct relationships/reasoning (1); clear labels/evidence or method (1). Matching counts for each atom type and a clear system boundary.

13. Choose one of the supplied data comparisons. Write a claim supported by those data, give the two values as evidence, and explain what further information would be needed for a causal explanation. (4 points)

Answer / scoring: Rubric (4): accurate limited claim (1); two correct data values (1); logical connection to claim (1); relevant limitation/next test (1). Accept varied claims bounded by the supplied data.

Social Studies - 24 points

Score: _____ / 24

W1 background: Sea and land routes connected trade, movement, and political control. Stories can express identity but are not automatically factual records. This is an assessment background note written for this packet, not an actual historical primary-source quotation.

W2 background: They can affect customs, roles, access, and expectations, with variation over time. Elite records may not represent the experience of all people. This is an assessment background note written for this packet, not an actual historical primary-source quotation.

W3 background: A republic uses institutions of public/representative rule; an empire rules over multiple peoples/territories, often with centralized authority; categories can overlap. Institutions and power changed across periods rather than remaining fixed. This is an assessment background note written for this packet, not an actual historical primary-source quotation.

W4 background: They supported movement, administration, trade, and water supply. No; interacting political, military, economic, and other factors matter. This is an assessment background note written for this packet, not an actual historical primary-source quotation.

1. How did the Mediterranean location support Rome's connections? (1 point)

Answer / scoring: Sea and land routes connected trade, movement, and political control.

2. Why distinguish founding traditions from historical evidence? (1 point)

Answer / scoring: Stories can express identity but are not automatically factual records.

3. Compare a founding story with material/written evidence. (3 points)

Answer / scoring: Rubric (3): accurate relevant content (1); supported comparison/relationship (1); clear organization and source/evidence awareness (1). Genre/purpose distinction, evidence to check, and a qualified conclusion.

4. How can religion and social status shape daily life? (1 point)

Answer / scoring: They can affect customs, roles, access, and expectations, with variation over time.

5. Why compare accounts from different social positions? (1 point)

Answer / scoring: Elite records may not represent the experience of all people.

6. Analyze a Roman daily-life account. (3 points)

Answer / scoring: Rubric (3): accurate relevant content (1); supported comparison/relationship (1); clear organization and source/evidence awareness (1). Creator/status/purpose, useful evidence, and missing perspectives.

7. How does a republic differ from an empire as broad political concepts? (1 point)

Answer / scoring: A republic uses institutions of public/representative rule; an empire rules over multiple peoples/territories, often with centralized authority; categories can overlap.

8. Why must Roman political changes be placed on a timeline? (1 point)

Answer / scoring: Institutions and power changed across periods rather than remaining fixed.

9. Compare Roman republican and imperial power. (3 points)

Answer / scoring: Rubric (3): accurate relevant content (1); supported comparison/relationship (1); clear organization and source/evidence awareness (1). Roles of institutions/leaders, time/context, limits on participation, and evidence.

10. Why were roads and aqueducts important to Roman power and life? (1 point)

Answer / scoring: They supported movement, administration, trade, and water supply.

11. Can one cause explain the fall of the western Roman Empire fully? (1 point)

Answer / scoring: No; interacting political, military, economic, and other factors matter.

12. Model a Roman legacy and a limitation of a single-cause explanation. (3 points)

Answer / scoring: Rubric (3): accurate relevant content (1); supported comparison/relationship (1); clear organization and source/evidence awareness (1). Accurate legacy and at least two interacting factors or competing explanations.

13. Choose one background note. Describe an additional perspective or source that could improve it, and explain what it might reveal or challenge. (4 points)

Answer / scoring: Rubric (4): identifies chosen note/claim (1); relevant additional source/perspective (1); explains new insight/check (1); recognizes limits rather than treating one source as complete truth (1).

Spanish - 12 points

Score: _____ / 12

Spanish model and vocabulary (cue support allowed)

Quiero el menú, por favor.
Word bank: el desayuno, el almuerzo, la cena, quiero, el menú, por favor.

1. Week 1: Translate bread into Spanish. (1 point)

Answer / scoring: pan / el pan

2. Week 2: Complete: Me ___ el arroz. (1 point)

Answer / scoring: gusta

3. Week 3: Translate dinner into Spanish. (1 point)

Answer / scoring: cena / la cena

4. Week 4: Translate dinner into Spanish. (1 point)

Answer / scoring: cena / la cena

5. Read the supplied Spanish model. Give its meaning in English. (2 points)

Answer / scoring: I want the menu, please.

6. Write one new correct sentence using vocabulary from this module. (2 points)

Answer / scoring: One point for relevant vocabulary; one for a meaningful correct sentence. Accept varied answers.

7. Oral task (teacher listens): Use 6-8 short statements or a 6-line exchange about this week/module topic. You may use the supplied vocabulary/model as a cue, but add your own relevant detail or question. (4 points)

Answer / scoring: Oral rubric (4): communicates topic meaning (1); uses relevant vocabulary (1); uses understandable phrases/sentences (1); responds to teacher or adds a relevant original detail/question (1). Sample cue: Quiero el menú, por favor. Accept different accurate answers and appropriate gender/person forms. Minor accent/pronunciation errors do not erase understood meaning.